Noise Covariance Defines Reconstruction Geometry: Optimal Filtering, Weighted Subspaces, and Tied Linear Autoencoders

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Abstract

Under a linear Gaussian measurement model the inverse noise covariance Σ^{-1} is the metric in which a detector reconstruction is learned; the architecture fixes only the admissible class and the training data the excited directions. Optimal filtering, covariance-weighted PCA, tied noise-aware linear autoencoders and their Kronecker-separable restriction are one Σ^{-1}-orthogonal projection differing only in class; the weighted residual predicts whitened, not Euclidean, clean-signal error. Equal-norm covariance errors need not act equally: to first order the learned subspace moves by the retained–discarded coupling block divided by the pairwise gaps, and diagonal error is harmless until it reorders the spectrum; equal-norm perturbations differed a hundredfold. The separable class's contrast with the unrestricted one splits exactly into a provably positive approximation cost and two estimation excesses; on fresh seeds it won at $N = 800$ and lost at $N = 1600$. Excitation in training did not make a physical direction recoverable. On seven CRESST channels covariance weighting lowered every held-out weighted residual (mean 21.6%); against injected TIDMAD truth no rank-2 whole-window reconstruction beat predicting zero, and the narrowband restriction that fixed that failure removed the metric's leverage with it.

1 Introduction

Scientific detectors record a structured response superposed with electronic, thermal and environmental fluctuations that are rarely independent and identically distributed: they are colored in frequency, heteroskedastic across samples, correlated between readout channels, or slowly varying in time. Reconstruction must therefore do more than smooth a waveform. It must avoid spending model capacity on directions dominated by detector noise.

Three communities solve closely related problems in different language. Signal processing projects a trace onto a known template with inverse-noise weighting [1, 2]; in cryogenic calorimetry this is the optimal-filter (OF) amplitude estimator [3–5]. Statistics replaces the fixed template by a learned low-dimensional model — probabilistic PCA under isotropic noise [6], factor analysis under diagonal noise [7], weighted or expectation–maximization PCA (EMPCA) under nonuniform weights [8, 9]. Machine learning writes the same projection as an encoder, a latent code and a

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decoder; under Euclidean mean-squared error a tied linear autoencoder recovers the PCA subspace [10–12].

The common object is the observation-space metric. For $x = s + n$ with $n \sim \mathcal{N}(0, \Sigma)$, the Gaussian negative log-likelihood of a candidate reconstruction $\hat{s}$ contains $(x - \hat{s})^\dagger \Sigma^{-1} (x - \hat{s})$, so Σ^{-1} decides which residual directions are costly. Isotropic MSE is not a neutral default: it is the Gaussian likelihood only when $\Sigma = \sigma^2 I$, and under colored noise it ranks residuals in a different geometry. A method can reduce raw MSE while worsening the covariance-weighted residual that follows from the assumed detector model. That is a likelihood misspecification, not a change of units.

The question. What determines which representation a detector model learns, and when does covariance weighting — or a structural restriction of the learned class — improve recovery of a physical signal? For a calibrated readout the theory exposes three determinants: the *metric* the measurement fixes, the *class* the architecture fixes, and the *support* the training data excite. This paper establishes the first two, states the third as an exact population identity, and then asks what each does at finite sample, where the covariance is estimated and the data are limited. The answers are mostly negative in the places where the finite-sample question bites, and the paper reports them that way.

Contributions.

(i) *One projection, four classes.* OF, covariance-weighted PCA, the tied noise-aware linear autoencoder and its Kronecker-separable restriction are one Σ^{-1}-orthogonal projection whose classes differ only in the admissible set, with the tied encoder as an adjoint condition and untying stated as a change of parameterization rather than of optimum (Section 3).

(ii) *Which risk the residual predicts.* The weighted residual predicts the population whitened clean-signal error of a fixed linear map exactly, up to a trace correction; it need not predict Euclidean clean-signal error, and a two-dimensional example separates them (Section 2.2).

(iii) *Covariance error acts locally.* To first order the learned subspace moves by the retained–discarded coupling block of the relative covariance error divided by the pairwise eigenvalue gaps (Proposition 3.2). A diagonal error in the fitted eigenbasis is exactly harmless until it reorders the spectrum; a near-singular estimate is a separate regime. In a controlled study, perturbations of equal norm differed by two orders of magnitude in effect, as the local statement predicts and the global norm bound cannot (Section 5.2).

(iv) *A structural prior trades approximation for estimation.* The finite-sample contrast between the separable and the unrestricted class decomposes exactly into a population approximation cost, for which a computable lower bound is given, and two estimation excesses (Proposition 3.3). On fresh seeds the restricted learner won at $N = 800$ and lost at $N = 1600$: both regimes were observed and the crossover is bracketed, not located (Section 5.3).

(v) *Excited is not detectable.* Scaling the excitation of one physical latent in the training draw did not set its recovery. All five registered predictions of that study failed; an exploratory reading in terms of the induced population spectrum and the direction's eigenvector loadings is recorded for an independent test (Section 5.4).

(vi) *The two determinants trade off on real data.* Against recorded TIDMAD truth no rank-2 whole-window reconstruction beat predicting zero. Restricting the same fit to a nine-bin band at the same window — a second predeclared hold-out, executed once — moved it below a no-denoising bin read, confirming the dimensional diagnosis in direction. But its dependence assessment did not stabilise, so all three registered verdicts are withheld, and the metric's contribution vanished with the band's dynamic range: the restriction that repairs estimation removes the geometry's leverage (Sections 6.4–6.5).

(vii) *An evidence programme with named boundaries.* For each public dataset we report which of three questions — reconstruction in the assumed geometry, calibration of the assumed covariance, agreement with recorded truth in a named metric — it can and cannot answer, including the failures (Table 2).

What is recalled and what is added. The identities organizing (i) are largely classical — that the generalized least-squares amplitude is the optimal filter, that covariance-weighted low-rank approximation reduces to an SVD of whitened data [13, 14], that Euclidean linear autoencoders recover PCA [10], and that alternating per-axis projections are multilinear PCA [15]. Statement (ii) specializes classical unbiased-risk accounting [16, 17] to a fixed linear map in whitened coordinates; (iii) is first-order invariant-subspace perturbation theory [18] applied to the operator the estimator actually diagonalizes; (iv) is the ordinary approximation–estimation decomposition of a constrained estimator with a Ky Fan bound. What is added is their statement in one geometry, and what the controlled programme found when it tested them. No statement here is claimed as an independently substantiated novelty; Supplement, Section S1 marks each as established, specialized, a corollary, a synthesis, or derived here.

Evidence arc. Controlled simulations first test consequences that are known under the data-generating model. Public CRESST-II/III pulse-shape data [19] then give a positive but narrow real-detector test of within-channel reconstruction, and public H1/L1 data around GW150914 [20] a negative calibration stress test. Neither release records the signal that was present, so both compare residuals rather than distance to truth; public TIDMAD data from the ABRACADABRA axion search [21] supply the missing ground truth through an injected reference channel. This order separates reconstruction performance from detection sensitivity, and implementation identity from statistical calibration.

One block of controlled planted-physics evidence for the separable class — whether the wrong metric learns the wrong physical structure, and how much of the excited span a finite sample recovers — is complete and is held from this version until the double-blind review of its companion workshop submission concludes on 29 September 2026. Nothing else is held; the September programme on the same planted family is reported in full (Section 5).

Scope. The scope is linear and Gaussian. The exact statements do not extend to nonlinear autoencoders, non-Gaussian transients, or a covariance model that is wrong for the evaluation data. CRESST is used only for within-channel reconstruction of public held-out pulses, not cross-channel transfer or detector sensitivity. GWOSC is used to test whether two specific PSD models calibrate a matched statistic on one public interval, not to estimate GW150914 significance. TIDMAD is used at named operating points on named validation files; it is not a survey of the release's parameter space. Detection sensitivity is not claimed anywhere. Full derivations, proofs, implementation

details, verification grids and per-file diagnostics are in the technical supplement, and every number traces to a checksummed record (Section 10).

2 Observation model and target risks

2.1 Model, whitening and the common objective

Write the measurement as

$$x_i = s_i + n_i, \qquad n_i \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma), \qquad \Sigma \succ 0, \qquad i = 1, \dots, N, \tag{1}$$

with $x_i \in \mathbb{C}^d$ a single trace (or a vectorized channel–time block) and s_i the structured response. The complex circular convention is used throughout; the real case is identical with $\dagger \to T$. Let $\Sigma = \Sigma^{1/2}\Sigma^{\dagger/2}$ and define the whitening map $y = \Sigma^{-1/2}x$, under which the noise becomes isotropic, $\Sigma^{-1/2}n \sim \mathcal{N}_{\mathbb{C}}(0, I)$.

Up to constants the Gaussian negative log-likelihood of a reconstruction $\hat{s}$ is

$$\chi^2(\hat{s}) = (x - \hat{s})^{\dagger}\Sigma^{-1}(x - \hat{s}) = \left\|\Sigma^{-1/2}(x - \hat{s})\right\|_2^2, \tag{2}$$

so every method in this paper solves one problem — minimize (2) — over a different admissible set $\mathcal{F}$ for $\hat{s}$. Fixing $\hat{s} \in \operatorname{span}\{p\}$ for a known template p gives the optimal filter; letting $\hat{s}$ range over a learned k-dimensional subspace gives covariance-weighted (EMPCA) reconstruction; parameterizing that subspace by an encoder and decoder gives the tied noise-aware linear autoencoder; restricting it to a Kronecker product of a channel and a time subspace gives the separable class. The measurement fixes the metric, the architecture fixes the class.

By contrast the isotropic loss $\mathcal{L}_{\mathrm{MSE}} = N^{-1}\sum_i \|x_i - \hat{x}_i\|_2^2$ coincides with (2) up to scale exactly when $\Sigma = \sigma^2 I$. The two losses define different inner products, hence different notions of which waveform directions matter: isotropic MSE weights every bin equally and is driven by raw variance, which under colored noise is typically dominated by noise-rich, low-information directions, while the Σ^{-1}-weighted loss is driven by signal-to-noise structure. The same class, rank, data and optimizer can therefore converge to substantially different subspaces. We call the resulting failure mode *metric reversal*: a method can achieve lower raw MSE together with a worse weighted residual — equivalently a worse held-out Gaussian likelihood — while learning a subspace nearly orthogonal, in the whitened metric, to the one efficient estimation requires.

2.2 Which risk the weighted residual predicts

The likelihood fixes a residual geometry. It does not by itself say which scientific error a reconstruction chosen in that geometry makes small, so every claim below names its target. For a reconstruction $\hat{x} = \hat{x}(x)$ of $x = s + n$ four risks are in play: the observation residual $\mathcal{R}_{\mathrm{obs}} = \mathbb{E}\,(x - \hat{x})^T\Sigma^{-1}(x - \hat{x})$; the whitened clean-signal risk $\mathcal{R}_{\mathrm{clean},\Sigma} = \mathbb{E}\,(s - \hat{x})^T\Sigma^{-1}(s - \hat{x})$; the Euclidean clean-signal risk $\mathcal{R}_{\mathrm{clean},I} = \mathbb{E}\,\|s - \hat{x}\|_2^2$; and the physical-parameter risk $\mathcal{R}_{\vartheta} = \mathbb{E}\,\|\hat{\vartheta} - \vartheta\|^2$.

Proposition 2.1 (Fixed affine-estimator risk identity)**.** *Work in real whitened coordinates, $y = \Sigma^{-1/2}x = t + w$. Let $\hat{y} = By + b$ be an affine reconstruction with (B, b) fixed — fitted on other data, or population quantities — and let the evaluation noise satisfy, conditionally on (B, b), $\mathbb{E}[w] = 0$, $\mathbb{E}[ww^T] = I$ and $\mathbb{E}[tw^T] = 0$. Then*

$$\mathcal{R}_{\mathrm{clean},\Sigma} = \mathcal{R}_{\mathrm{obs}} + 2\,\mathrm{tr}(B) - d. \tag{3}$$

Every idempotent B of rank k — orthogonal or oblique — has $\operatorname{tr} B = k$ and correction $2k - d$; a shrinkage estimator $B = \sum_j \beta_j u_j u_j^T$ has correction $2\sum_j \beta_j - d$.

The proof is two lines of unbiased-risk accounting [16, 17] and is in Supplement, Section S2. Four consequences bound what the rest of the paper may claim. *(a)* The identity is a *population* statement conditional on the fitted (B, b); its moment conditions must hold for the evaluation noise given the training data. A chronological split makes them plausible but does not establish them under dependent noise, and a wrong Σ violates $\mathbb{E}[ww^T] = I$ and biases the correction. *(b)* Within any set of fixed estimators of *equal trace* — including all rank-k idempotent maps, so a raw-PCA projector expressed in true whitened coordinates is covered — ordering by the observation residual is ordering by the whitened clean-signal risk. That is the precise sense in which the covariance-weighted subspace is the better reconstruction; on a finite sample the two orderings can differ, so every comparison below carries its own interval. *(c)* The identity connects a *held-out* residual to the clean-signal risk of the fitted map, and says nothing about the training optimum. *(d)* It is silent about other targets: maps of unequal trace need the correction, a nonlinear or data-dependent map is outside it, and neither the Euclidean clean-signal risk nor a physical-parameter risk is implied. Training on a clean target with isotropic MSE is therefore not an assertion that the noise is white — it is the choice of $\mathcal{R}_{\text{clean},I}$ as the prediction risk, a legitimate target the weighted residual need not predict.

Example 2.2 (The two clean-signal risks can order oppositely)**.** *Let $d = 2$, $k = 1$, with independent zero-mean Gaussian signal and noise, $\mathbb{E}[ss^T] = \operatorname{diag}(100, 30)$ and $\Sigma = \operatorname{diag}(4, 1)$. Raw PCA retains e_1 (observed variances $104, 31$); weighted PCA retains e_2 (whitened observed variances $26, 31$). With the exact covariance, infinite data and exact optimization, $\mathcal{R}_{\text{obs}}$ and $\mathcal{R}_{\text{clean},\Sigma}$ are 31 against 26 — both prefer weighted PCA, as Proposition 2.1 requires with $2k - d = 0$ — while $\mathcal{R}_{\text{clean},I}$ is 34 against 101 and prefers raw PCA, because the Euclidean clean error of the projector onto e_j is the signal variance it discards plus the noise variance it keeps. The example separates the objective from another legitimate risk: a disagreement between the two orderings is not by itself evidence for the likelihood-aligned one.*

2.3 Three testable predictions

Holding class, rank, data, preprocessing and split fixed and changing only the loss geometry gives three predictions, which organize the evidence of Sections 5–6: **(P1)** under white noise, isotropic PCA and EMPCA recover the same subspace and agree on all metrics — an internal control, since if the two coincide at $\Sigma \propto I$ and separate under colored noise, the effect is attributable to loss geometry rather than to rank, optimization or preprocessing; **(P2)** under colored noise the objectives diverge, each winning in the metric it optimizes; **(P3)** where a clean signal or injected reference is recorded, the reconstruction preferred by the weighted residual also has the lower clean-signal error in a named metric. For the whitened clean-signal risk and fixed maps of equal trace, (P3) is the population content of Proposition 2.1; for the Euclidean risk it is not implied (Example 2.2); for a physical parameter scored against a reference — what the TIDMAD stage measures — it is an empirical hypothesis of this paper. (P3) is what distinguishes the claim from a tautology, because (P2) alone says only that each objective wins in its own metric. (P1) and (P2) are testable on any release; (P3) needs a dataset that records the injected signal.

A gain in the weighted residual is not by itself a gain in energy resolution or detector sensitivity, which additionally require a frozen signal-injection and background protocol; none is claimed here.

2.4 Pulling the metric back onto a representation

Two further consequences of the same metric fix what the controlled evidence can measure; both are stated and proved in Supplement, Section S3. *Pullback*: for a forward model $x = g(\vartheta) + n$ the Fisher information of the physical latents is $J_g^T \Sigma^{-1} J_g$, so the likelihood pulls the Σ^{-1} inner product back onto latent space and a latent displacement's length is the distance it moves the observation *in units of what the noise can resolve*; the same construction applies to any representation whose prediction is compared with x under Σ, and a Euclidean latent distance is the wrong statistic in the same sense that isotropic MSE is, coinciding with it only where the pulled-back form is a multiple of the identity. *Excited support*: if the training distribution varies only the latent coordinates S, the population rank-k objective depends on the projector only through $\mathrm{tr}(\Pi R_t)$ and is *exactly* invariant under every rotation fixing the excited span pointwise; with $r = \operatorname{rank} R_t$, the minimizers contain $\operatorname{range} R_t$ when $k \geq r$ and leave the remaining $k - r$ dimensions undetermined, and are its top-k eigenspaces when $k < r$. Unexcited directions are invisible to the population objective, and a finite sample recovers the excited span only above its own sampling-noise floor.

The three determinants are now explicit: the measurement fixes the metric Σ^{-1}, the architecture fixes the class $\mathcal{F}$, and the training data fix the excited support S. Sections 3 and 4 establish the first two; Section 5 asks what each does at finite sample.

3 Recovery theory: one projection, four classes

3.1 The hierarchy

Every estimator in this paper minimizes (2) over a different admissible set. Table 1 is the whole hierarchy; Figure 1 draws it as one projection whose faces are transposes. The classes are nested, OF $\subset$ NFPA $\subset$ EMPCA, with the tied linear autoencoder a *parameterization* of the EMPCA class rather than a fifth class.

Table 1: One Σ^{-1}-orthogonal projection, four admissible classes. C channels, T time samples, $d = CT$. "Degenerates to" names the exact finite-sample coincidence, not an asymptotic one.

class	admissible $\hat{s}$	free parameters	degenerates to
optimal filter	αp, p known	$\alpha \in \mathbb{C}$	—
EMPCA (rank k)	$\hat{s} \in \mathcal{S}$, $\dim \mathcal{S} = k$	Grassmannian $\mathrm{Gr}(k, d)$	OF at $k = 1$, $p \in \mathcal{S}$
tied linear AE	as EMPCA, $P^\dagger \Sigma^{-1} P = I_k$	$P \in \mathbb{C}^{d \times k}$	EMPCA exactly (Thm. 3.1)
NFPA (separable)	$\mathcal{S} = \mathcal{S}_c \otimes \mathcal{S}_t$	$\mathrm{Gr}(k_c, C) \times \mathrm{Gr}(k_t, T)$	EMPCA iff the optimum is separable; OF at $(1, 1)$

3.2 The bridge to representation learning

Theorem 3.1 (Bridge Theorem: noise-aware linear AE $\equiv$ EMPCA)**.** *Let $x_1, \ldots, x_N \in \mathbb{C}^d$ be any observations, $\Sigma \succ 0$ any Hermitian weight, $1 \leq k \leq d$. Consider the tied noise-aware loss*

$$\mathcal{L}(P) = \frac{1}{N} \sum_{i=1}^{N} (x_i - PP^\dagger \Sigma^{-1} x_i)^\dagger \Sigma^{-1} (x_i - PP^\dagger \Sigma^{-1} x_i), \qquad P^\dagger \Sigma^{-1} P = I_k, \tag{4}$$

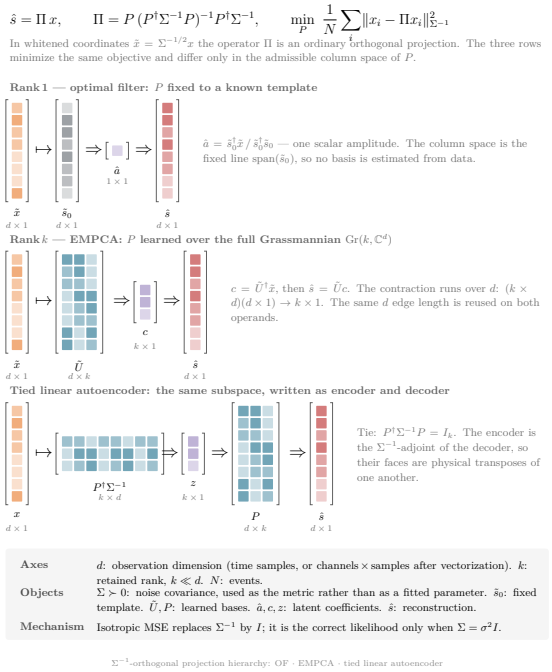


Figure 1: The estimators as one Σ^{-1}-orthogonal projection, differing only in the admissible column space of P. Rank-1 optimal filtering fixes that column space to a known template; EMPCA learns a k-dimensional subspace over the full Grassmannian; the tied linear autoencoder writes the same subspace as an encoder and a decoder related by $P^\dagger \Sigma^{-1} P = I_k$, so the two faces are transposes. Block widths and heights encode matrix shapes; cell counts illustrate axis structure and are not literal dimensions.

and let $\lambda_1 \geq \cdots \geq \lambda_d \geq 0$ *be the eigenvalues of the whitened second moment* $\Sigma^{-1/2}\hat{R}_x\Sigma^{-1/2}$*. Then* P^* *minimizes* $\mathcal{L}$ *if and only if* $\Sigma^{-1/2}P^*$ *spans a top-k invariant subspace of that operator — equivalently,* P^* *spans the top-k generalized eigenspace of* $\hat{R}_x\Sigma^{-1}v = \lambda v$ *— and* $\mathcal{L}^* = \operatorname{tr}(\Sigma^{-1}\hat{R}_x) - \sum_{j\leq k}\lambda_j$*. If* $\lambda_k > \lambda_{k+1}$ *the subspace is unique and equals the rank-k EMPCA range, up to a unitary rotation within it. If* $\lambda_k = \lambda_{k+1}$ *the minimizers form a family; in particular when* $\operatorname{rank}\hat{R}_x < k$ *every direction of* $\Sigma\ker\hat{R}_x$ *— the null space of* $\hat{R}_x\Sigma^{-1}$*,* not *of* $\hat{R}_x$ *— is an equally good completion.*

Whitening turns the constraint into $W^\dagger W = I_k$ and the loss into $N^{-1}\|\tilde{X}(I - WW^\dagger)\|_F^2$, so Rayleigh–Ritz gives the result; the proof is in Supplement, Section S4.

Remark 3.1 (What the theorem uses, and what it does not)**.** *The bridge is a* deterministic identity *between two optimization problems: it holds for any data, any* $\Sigma \succ 0$ *and any* $N \geq 1$*, and its proof uses no noise model — the measurement model enters only in motivating the weight. It characterizes the* global *minimizers, whereas the alternating algorithm reaches a stationary point and every k-dimensional invariant subspace is stationary, so agreement with the global optimum is a property of the initialization and the spectrum. The minimizer is unique only up to the latent gauge* $P^* \mapsto P^*Q$*; physical quantities depend on the reconstruction and are gauge-invariant. And* untying *the encoder changes neither the optimal reconstruction nor the class — only the parameterization and the landscape (Supplement, Section S5).*

3.3 The metric is architecture-neutral

The weighted loss is not one modeling choice among many. For quadratic residual losses $\mathcal{L}_M(r) = r^\dagger Mr$ with $M \succ 0$, invariance under the isometry group of the measured noise — the maps preserving Σ — is *equivalent* to $M \propto \Sigma^{-1}$ (Supplement, Section S6). The argument acts on the residual vector alone and mentions no architecture, so a reconstruction method is the pair $(\Sigma^{-1}, \mathcal{F})$ of a residual geometry fixed by the likelihood and an admissible class fixed by the architecture. This is what makes "metric" and "class" separable determinants rather than two names for the same choice.

A related invariance is worth stating because it is often assumed rather than checked: the projector $PP^\dagger\Sigma^{-1}$ and every reconstruction derived from it are invariant under the joint rescaling $(\Sigma, x) \mapsto (c^2\Sigma, cx)$, so nothing in the hierarchy depends on the physical units of the readout; only the relative shape of Σ enters.

3.4 Covariance error acts locally

In practice Σ is estimated. Write the supplied covariance as $\hat{\Sigma} = \Sigma^{1/2}(I + E)\Sigma^{1/2}$ with E Hermitian and $\varepsilon = \|E\|_{\mathrm{op}} < 1$; ε is the natural *relative* error, invariant under the rescaling above. A Davis–Kahan argument gives the global bound $\sin\Theta \leq 6\varepsilon\lambda_1/\Delta_\lambda$ for $\varepsilon \leq 1/2$ (Supplement, Section S7). That bound is valid but, where it applies, usually uninformative: on the planted two-axis configuration of Section 5.3 it is vacuous at any ε of practical size, because $\lambda_1/\Delta_\lambda \approx 2.7 \times 10^3$ at rank six. The local statement is the informative one.

Proposition 3.2 (First-order displacement under covariance error)**.** *Let* A *be the whitened operator whose top-k eigenspace the estimator returns, with eigenpairs* (λ_i, q_i)*, retained set* $\mathcal{R} = \{1, \ldots, k\}$ *and* $\lambda_k > \lambda_{k+1}$*; let* $E_{ji} = q_j^T E q_i$*; and let* $P, \hat{P}$ *be the projectors, in the whitened coordinates of the*

true Σ, onto the subspaces fitted with Σ and with $\hat{\Sigma}$. Then $\hat{P} = P + \delta P + O(\varepsilon^2)$ with

$$q_j^T\, \delta P\, q_i = -\frac{\lambda_j E_{ji}}{\lambda_i - \lambda_j} \qquad (i \in \mathcal{R},\ j \notin \mathcal{R}), \tag{5}$$

and every other entry of δP zero. Consequently: (1) *only the coupling block $E_{\mathcal{R}^\perp\mathcal{R}}$ rotates the subspace to first order — the within-retained and within-discarded blocks deform coordinates and rotate nothing;* (2) *each coupling entry is weighted by the discarded eigenvalue and divided by its* own *pairwise gap, not by the global gap, so $\|\delta P\|_F^2 = 2\sum_{i\in\mathcal{R},j\notin\mathcal{R}} \lambda_j^2 E_{ji}^2/(\lambda_i - \lambda_j)^2$;* (3) *if E commutes with A (entries $e_i > -1$) the statement is exact at every order — the eigenvectors are unchanged, the eigenvalues become $\lambda_i/(1+e_i)$, and the fitted subspace equals the oracle one until the perturbed spectrum reorders, at which point the retained set is exchanged;* (4) *if the estimator diagonalizes a sample operator $\hat{A} = A + D$, the joint first-order displacement from the population subspace is $(D_{ji} - \lambda_j E_{ji})/(\lambda_i - \lambda_j)$ — the training-sample and covariance terms add and are separated by the same gaps. Expanding instead in the eigenbasis of $\hat{A}$, at fixed training data, removes D and gives* (5) *with $(\hat{\lambda}_i, \hat{q}_i)$; that is the form the controlled test of Section 5.2 checks. The remainder is of order $\varepsilon^2\lambda_1(A)^2/\delta^2$ with $\delta = \min_{i\in\mathcal{R},j\notin\mathcal{R}} |\lambda_i - \lambda_j|$, so the first-order statement is informative when $\varepsilon\lambda_1/\delta \ll 1$, not merely when $\varepsilon \ll 1$.*

The expansion requires $I + E \succ 0$. A near-singular $\hat{\Sigma}$, $\lambda_{\min}(I+E) \to 0$, is *outside* it, and outside the global bound as well; Section 5.2 reports that regime separately. A covariance model wrong in *structure* rather than merely noisy need not have a large ε — $\hat{\Sigma} = \Sigma^{1/2}(I + \delta vv^T)\Sigma^{1/2}$ is a covariance with $\varepsilon = |\delta|$ — but its ε does not vanish as the calibration sample grows: the estimate converges to the misspecified model. That case is what the GWOSC stage probes. The proof is in Supplement, Section S7.

3.5 A structural prior trades approximation for estimation

Holding the metric fixed, the fourth class *restricts* the learned subspace: for a two-axis measurement $Z \in \mathbb{C}^{C\times T}$ with separable noise $\Sigma = \Sigma_c \otimes \Sigma_t$, the noise-factored projection autoencoder (NFPA) admits only $\mathcal{S}_c \otimes \mathcal{S}_t$. The restricted bridge (Supplement, Section S8) says it solves the same objective over a product Grassmannian, with equality to the unrestricted optimum exactly when that optimum is separable; at $(k_c, k_t) = (1,1)$ with a separable template it degenerates to the joint-channel optimal filter. The separable class matters because it is the first at which the *architecture*, not the metric, decides what can be represented.

Its cost is measurable, and the following identity is what a finite-sample class contrast actually measures.

Proposition 3.3 (Exact risk decomposition and a lower bound on the restriction cost)**.** *Let $K \succeq 0$ be the whitened clean-signal second moment with eigenvalues $\mu_1 \geq \mu_2 \geq \cdots$; for a rank-$k$ projector P ($k = k_c k_t$) let $R_K(P) = \operatorname{tr} K - \operatorname{tr}(PK)$; let $P_U^\star$ be the top-k eigenprojector of K with gap $\Delta_k = \mu_k - \mu_{k+1}$; let $\mathcal{F}$ be the compact set of product projectors and $P_\mathcal{F}^\star \in \arg\min_\mathcal{F} R_K$. Then for any fitted $\hat{P}_\mathcal{F} \in \mathcal{F}$ and any fitted rank-k projector $\hat{P}_U$,*

$$R_K(\hat{P}_\mathcal{F}) - R_K(\hat{P}_U) = \underbrace{R_K(P_\mathcal{F}^\star) - R_K(P_U^\star)}_{a\ \geq\ 0} + \underbrace{R_K(\hat{P}_\mathcal{F}) - R_K(P_\mathcal{F}^\star)}_{x_\mathcal{F}\ \geq\ 0} - \underbrace{R_K(\hat{P}_U) - R_K(P_U^\star)}_{x_U\ \geq\ 0}, \tag{6}$$

an identity, and for every $P_F \in \mathcal{F}$

$$R_K(P_F) - R_K(P_U^\star) \;\geq\; \Delta_k\big(k - \mathrm{tr}(P_F P_U^\star)\big), \quad \mathrm{tr}(P_F P_U^\star) \leq \min\Big\{k,\ \textstyle\sum_{i\leq k_c} \mu_i(M_c),\ \sum_{i\leq k_t} \mu_i(M_t)\Big\}, \tag{7}$$

where M_c *and* M_t *are the partial traces of* $P_U^\star$ *over the temporal and channel factors. Hence* a *is bounded below by a quantity computed from* K *alone, and when* $\Delta_k > 0$ *it vanishes if and only if* $P_U^\star \in \mathcal{F}$.

The bound is what makes the sign of the first term a fact about the generator rather than about the fits. Two cautions govern its use. An observed signed contrast between held-out weighted residuals at matched rank is a *finite-sample class contrast*: it mixes a with the two estimation excesses, and it can be negative — the restriction acting as a regularizer at finite N — without the population approximation error being negative, which it cannot be. And agreement of the objective across random restarts bounds the optimization term *only for the restarts tried*; it does not certify a global optimum. Section 5.3 reports both terms under those names. The proof is in Supplement, Section S9.

4 Estimator instances

4.1 Optimal filtering as the rank-1 case

Fixing $\hat{s} = \alpha p$ for a known template p turns (2) into a one-parameter generalized least-squares problem with the closed-form minimizer and variance

$$\hat{\alpha} = \frac{p^\dagger \Sigma^{-1} x}{p^\dagger \Sigma^{-1} p}, \qquad \mathrm{Var}(\hat{\alpha}) = \frac{1}{p^\dagger \Sigma^{-1} p}, \tag{8}$$

which attains the Cramér–Rao bound for the amplitude under the assumed model. For stationary noise with one-sided PSD J, the kernel of (8) is the familiar $\tilde{p}_k^*/J_k$ matched filter and the resolution is $\sigma_E^{-2} = \sum_k |\tilde{p}_k|^2/J_k$. Time-shift and joint-channel OF are the same projection with an enlarged fixed class (Supplement, Section S10); the estimator is unbiased only if the true response lies in $\mathrm{span}\{p\}$, and template mismatch biases it by the component of the response orthogonal to p in the Σ^{-1} metric — the limitation the learned classes remove.

The token-axis principle. For wide-sense stationary noise periodic with the record length, Σ is circulant and $\Sigma^{-1} = F^\dagger \,\mathrm{diag}(1/J_k) F$: the eigenvectors of the precision operator *are* the discrete Fourier modes and its eigenvalues are the inverse-PSD weights of the OF kernel, so an encoding of the time axis by the leading eigenvectors of Σ^{-1}, weighted by their eigenvalues, is the matched-filter weighting itself. This is the construction positional encodings for attention arrive at from the other side, where the operator is an *assumed* graph Laplacian [22–24]; here it is the *measured* noise precision. Three qualifications keep it where it belongs, and are spelled out in Supplement, Section S10: the precision weights $1/J_k$ are not the whitening scalings $1/\sqrt{J_k}$, so the proposition identifies a basis and its weights and not a preferred encoding; nothing here concerns a learned model, since a learned query–key projection changes the geometry attention computes; and on a finite non-periodic window the covariance is Toeplitz, asymptotically equivalent to its circulant counterpart in normalized Hilbert–Schmidt norm but *not* in operator norm [25], so the exact statements here need the periodic model.

4.2 Covariance-weighted subspaces

Letting $\hat{s}$ range over a learned rank-k subspace gives

$$\min_{U,\,\{c_i\}} \ \frac{1}{N}\sum_i (x_i - Uc_i)^\dagger \Sigma^{-1}(x_i - Uc_i), \tag{9}$$

whose solution in whitened coordinates is the top-k SVD subspace of $\Sigma^{-1/2}X$ — a classical reduction [13, 14] implemented here by the EMPCA alternation [8]. Two population statements follow. At $k = 1$ with a unique leading whitened signal eigendirection and nonzero eigengap, weighted PCA recovers the OF direction; at finite sample the degeneracy is exact whenever the fitted rank-1 basis spans the template, in which case the two estimators coincide identically, not merely in expectation. At $k > 1$ containment is *not* automatic: the learned subspace need not contain the template, and the OF–EMPCA comparison therefore tests the benefit of enlarging the class, while the same-rank PCA–EMPCA comparison tests the contribution of the geometry. We keep the two apart throughout, because a gain over a fixed template is not by itself evidence that inverse-covariance weighting caused it.

The subspace is determined only up to the gauge $U \mapsto UQ$; latent coordinates are not observables, and every quantity reported below is a function of the projector or of the reconstruction. Finite-sample stability is governed by the whitened spectral gap through a Davis–Kahan argument, made local and quantitative by Proposition 3.2. Rank selection is not part of the theory: rank is fixed by the protocol in every experiment below and is never selected on the evaluation data.

4.3 Tied and untied autoencoders

Writing the same projection as an encoder and a decoder makes the connection to representation learning explicit. With decoder P and encoder $P^\dagger\Sigma^{-1}$ — the Σ^{-1}-adjoint of the decoder, which is the tying condition — the reconstruction is $PP^\dagger\Sigma^{-1}x$ and Theorem 3.1 identifies its optimum with EMPCA's. Untying the encoder leaves the optimal reconstruction and the class unchanged; it changes the parameterization and the optimization landscape, and in a controlled comparison at fixed budget the untied arm needed more iterations without finding a better optimum (Supplement, Section S11). The practical consequence for loss design is direct: a noise-aware autoencoder trained under (2) is not a heuristic reweighting but the Gaussian likelihood of the assumed detector model, and its optimum is a classical weighted-PCA subspace.

4.4 The separable class and its alternation

For $Z \in \mathbb{C}^{C\times T}$ with $\Sigma = \Sigma_c \otimes \Sigma_t$, per-axis whitening $\tilde{Z} = \Sigma_c^{-1/2} Z\, \Sigma_t^{-\dagger/2}$ turns the objective into a Frobenius one, and NFPA restricts the reconstruction to $U_c U_c^\dagger \tilde{Z}\, U_t U_t^\dagger$ with orthonormal factors. The restricted bridge makes this the same objective over $\mathrm{Gr}(k_c, C)\times\mathrm{Gr}(k_t, T)$, solved by alternating per-axis eigendecompositions — multilinear PCA [15] in the whitened metric. The parameter count drops from $k(CT-k)$ to $k_c(C-k_c)+k_t(T-k_t)$, which is the source of the estimation saving that Proposition 3.3 sets against the approximation cost. The alternation is not convex: agreement across restarts bounds the optimization term only for the restarts tried, and every claim below that depends on it is labelled *best-found* rather than optimal. Gauge and update details are in Supplement, Section S12.

5 Controlled evidence

5.1 Conventions

Every study below was predeclared, executed once, and is reported with the registered predictions that were *not* met. Rank is fixed by the protocol and never selected on evaluation data; splits are chronological or seed-paired as stated; intervals are bootstrap unless named otherwise, and a 95% interval is reported beside every contrast. The programme file, the protocol JSONs and the per-task fit caps are archived, and every number traces to a checksummed record (Section 10). The full verification grid, per-cell tables and diagnostics are in Supplement, Section S13–S16.

Nine frozen regression gates (S1–S9) verify the estimator identities rather than the paper's scientific claims; they are tabulated in full in Supplement, Section S13. Three carry the predictions of Section 2.3. Under white noise PCA and EMPCA differed by 1.48×10^{-6} degrees in maximum principal angle — (P1), at the arccos precision floor. An independently trained tied linear autoencoder reached the EMPCA objective with mean relative gap 1.95×10^{-16}, confirming Theorem 3.1 numerically. And on the colored-noise gate S5, over 30 seeds, the paired PCA-minus-EMPCA held-out weighted residual was $+0.550$ $[0.530, 0.571]$ and the held-out negative log-likelihood $+11.67$ $[11.24, 12.11]$, both favouring EMPCA, while *raw MSE preferred the opposite reconstruction* (2330 against 7413) and the two subspaces were nearly orthogonal in the whitened metric ($86.25°$) — (P2), with (P3) holding for the Euclidean clean-signal error in all thirty seeds (58.2 against 5110.9), which by Example 2.2 is a property of this generator and not a guarantee.

5.2 Covariance error: what actually moves the subspace

Two studies on one frozen generator ($d = 256$, exponent -1 power-law PSD, rank-2 signal at SNR 3 and 3/2, $N = 512$, $\lambda_1/\Delta_\lambda = 3.74$) ask how a wrong covariance moves the learned subspace.

The first (T15, 80 fits) finds the global bound valid on all 80 fits but *uninformative where its hypothesis holds* ($6\varepsilon\lambda_1/\Delta_\lambda$ is 8.6 and 4.3, above the value 1 that $\sin\Theta$ cannot exceed), and produces one instructive pair at nearly equal ε: a structured correlation-length misspecification ($\varepsilon = 0.557$) is nearly harmless ($\sin\Theta = 0.012$), while an empirical covariance from $m = 256 = d$ traces ($\varepsilon = 0.575$) is catastrophic ($\sin\Theta = 0.94$). An upper bound in ε never implied that perturbations of equal ε act alike; what this shows is that ε is an *incomplete summary*. Arm-by-arm results are in Supplement, Section S14.

The second study (T19, 38 arms, ten seeds, 380 fits) tests the local statement that explains why. Because the fit diagonalizes the *sample* whitened operator, the perturbations were engineered in the sample eigenbasis at a fixed training draw, so the training-sample term drops out and the displacement isolates the covariance term. Three regimes appeared (Supplement, Section S14, Fig. S1). *Coupling*: on the designed coupling, random-symmetric and stationary/Toeplitz families at $\varepsilon \le 0.2$ the measured Frobenius displacement was within a factor of two of the first-order value of Proposition 3.2 on 70 of 70 fits (medians 0.98–1.02), with residuals scaling as $\varepsilon^{2.05}$ for a generic E and $\varepsilon^{2.9}$ for a pure coupling block, whose second-order term vanishes identically; the registered slope prediction for the coupling family is not something the theory implies and was scored not met. *Separation*: a diagonal E preserving the eigenvalue ordering left the subspace unchanged ($\sin\Theta \le 4 \times 10^{-8}$ on 40 of 40 fits up to $\varepsilon = 0.4$), and an order-reversing one exchanged the retained subspace exactly when the *sample* spectrum check flipped (30 of 30; sample threshold 0.35–0.43 against the population value 0.72). *Conditioning* is a separate regime, outside both the proposition

and the global bound: an empirical covariance from $m \le d$ traces has $\lambda_{\min}(I + E) < 0.05$ on all 20 fits; at $m = d$ the subspace is destroyed ($\sin\Theta = 0.99$), at $m = d/2$ with a uniform eigenvalue floor it is only partially lost (median 0.73), and NLL-selected shrinkage restores $\lambda_{\min}$ to 0.21–0.35 while leaving $\sin\Theta$ between 0.21 and 0.99 — the regularizer repairs the spectrum floor, not the direction of the sampling error.

This is a *conditional* check at fixed training data with engineered perturbations, not a test of an independently estimated covariance, and it is a local theorem check, not a finite-perturbation guarantee. A finite-perturbation bound of this form remains open.

5.3 A structural prior at finite sample

Proposition 3.3 says what the separable restriction costs and buys. Two studies on a planted two-axis family measure it.

T14 first mapped the finite-sample class contrast at $C = 8$, $T = 256$, $(k_c, k_t) = (2, 3)$ over $N \in \{200, \dots, 3200\}$ and $\alpha \in \{0, 0.4, 0.8\}$ (25 cells, 1000 fits): the contrast was negative in every cell at SNR 10 — the restriction helping, with paired intervals excluding zero in fourteen of fifteen — and *no crossover was found anywhere on the grid*, so the predeclared confirmation stage was not triggered and the original gate, a transition within the grid, was not met (Supplement, Section S15). Because the population approximation cost at $\alpha > 0$ was never established independently of the fits, the follow-up measured it directly.

T20 carried out the population accounting at $T = 32$ ($d = 256$), same ranks and SNR 10, with K estimated from 2×10^5 clean events per α. The unrestricted reference is the top-six eigenprojector of K; the restricted reference is the *best of 51* alternating least-squares solutions on K, all agreeing to 10^{-13} — restart agreement, not a certificate. The resulting cost a is -7×10^{-15} at $\alpha = 0$, where the product family contains the excited signal exactly (the planted rank-six class excites only four directions there, because the coefficient map ties two entries), and 2.71 and 2.82 at $\alpha = 0.4$ and 0.8, about 3% of $\operatorname{tr} K$. The lower bound (7) gives 0.10 and 0.55: *the cost is provably positive at both nonzero* α, and the true optimum lies between the bound and the best-found value. Both apply to the archived Monte-Carlo covariance proxy, not to an exactly known population.

Over $N \in \{50, \dots, 1600\}$ (180 paired comparisons) the identity holds by construction, $x_U > x_{\mathcal{F}}$ throughout with c_U/c_F between 3 and 5, and the *registered c/N form is refuted*: the unrestricted excess decays with effective slopes $N^{-0.49}$ and $N^{-0.38}$, and the crossover those constants forecast near $N = 160$ is not there. On fresh seeds 10–19, chosen after the exploration and evaluated once, the contrast is negative at $N = 800$ (-0.39 $[-0.49, -0.29]$ and -0.29 $[-0.40, -0.18]$) and positive at $N = 1600$ ($+0.90$ $[+0.83, +0.96]$ and $+0.87$ $[+0.80, +0.94]$) at $\alpha = 0.4$ and 0.8: the restricted learner wins at $N = 800$ and the unrestricted one at $N = 1600$ for both tested nonseparabilities, while at $\alpha = 0$, where $a = 0$, it is negative at every N. An exploratory log-linear interpolation places the sign change near $N = 960$; that is interpolation, the registered $1/N$ extrapolation did not predict it, and $N^\star$ is *bracketed, not located*. The T14 null at $T = 256$ stands as a null: the configurations differ in dimension and pulse family, and the whitened spectra at the three α were not matched as the protocol asked, so within-α comparisons across N are the evidence. Restart selection by training loss matched the population-best restart in 50–90% of fits, with spreads up to 0.18 — of the order of the contrast near the sign change — so the optimization term is visible exactly where the classes are closest.

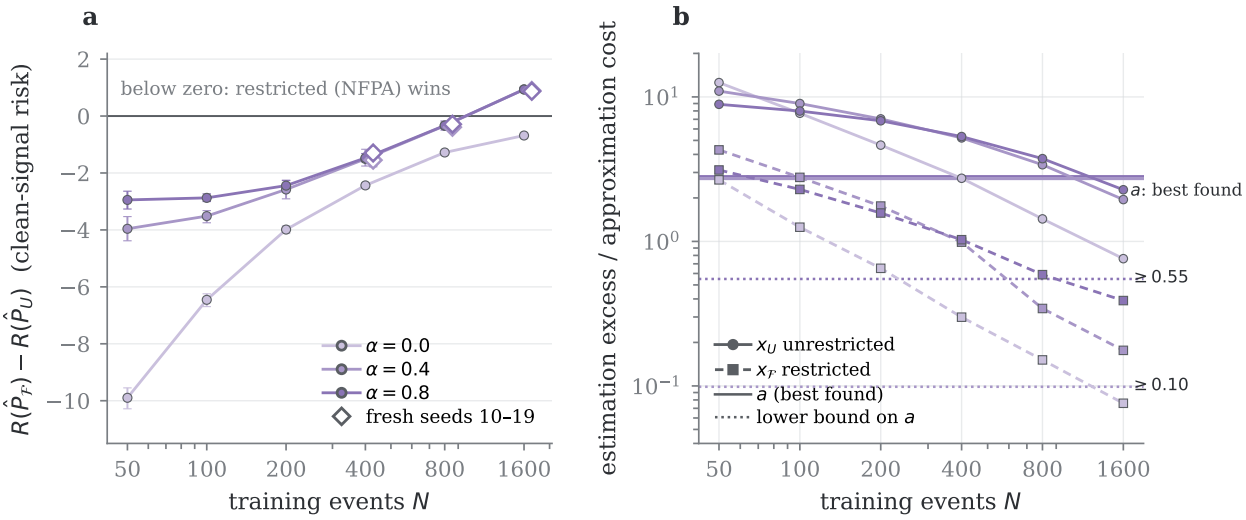


Figure 2: Structural prior at finite sample (T20; planted two-axis family, $C = 8$, $T = 32$, $(k_c, k_t) = (2, 3)$, SNR 10, oracle covariance). (a) Clean-signal risk contrast between the fitted separable and unrestricted subspaces against N: exploration seeds 0–9 with seed-paired 95% intervals, and fresh seeds 10–19 evaluated once (open diamonds). (b) The terms of Proposition 3.3: the estimation excesses x_U (solid) and $x_{\mathcal{F}}$ (dashed), the best-found approximation cost a (solid horizontal) and its provable lower bound (7) (dotted). The contrast changes sign where $x_U - x_{\mathcal{F}}$ falls through a. Registered inputs: `generated/SOURCE_MAP_part2.md`.

5.4 Excited is not detectable

Section 2.4 covers two cases — a direction the training data excite and one they do not. T21 scaled the excitation of one physical latent continuously between them (planted family at $\alpha = 0$, $C = 8$, $T = 256$, oracle covariance, SNR 10, $s \in \{0, 0.03, 0.1, 0.3\}$ with the others at 1, plus three controls; 19 cells, ten seeds, 190 closed-form rank-4 fits) and asked what sets recovery $\|Pu_j\|^2/\|u_j\|^2$ of the whitened Jacobian column u_j.

None of the five registered predictions was met as registered. Recovery is not a function of the excitation ratio alone. At $s = 0$ recovery is 0.13–0.37 for three of four directions against a geometric chance of 0.002, because the planted directions are not orthogonal (Gram entries to -0.65); the $s = 0$ probe returning the prior is a constant-target artefact, not evidence about the representation. The weak-regime variance signature failed under its registered baseline, the 0.95 strong-excitation level was not reached at $N = 800$ and only two directions reached it at $N = 3200$, and the across-seed recovery variance is *peaked* at the detection edge rather than monotone.

An exploratory reading — post hoc, recorded for an independent test, claimed as nothing more — is that recovery is set by the spike strengths of the population signal covariance relative to d/N and by each direction's loadings on the population eigenvectors: the fourth population spike, 0.54 above the noise floor, lies below $\sqrt{d/N}$ at both N, and the isolated-spike overlap combined with those loadings reproduces all eight observed full-excitation recoveries to within 0.01. The grid lesson is recorded too: $s \in \{0.03, 0.1\}$ are near-replicas of $s = 0$ because the induced variance scales as s^2, so a successor should grid on eigen-spike strength, keep the loadings, measure each direction's unique component, use a frozen readout and include a genuine all-zero control. *Excited and detectable are different properties of a planted direction*, and no abstention threshold for any

downstream diagnostic is established here.

6 Instrument evidence

6.1 Three instruments, three questions

No public release answers every question the theory raises, and the three used here answer different ones. Table 2 states, for each release, which of three questions it can and cannot decide: whether the reconstruction improves *in the assumed geometry*; whether the *assumed covariance is calibrated* on that data; and whether the reconstruction agrees with *recorded truth in a named metric*. Detection sensitivity is not among them and is not claimed anywhere in this paper.

Table 2: What each public release can and cannot answer. "–" marks a question the release cannot pose, not a test that failed. Untested quantities are named as such; detection sensitivity is not claimed for any release.

	Q1. Fixed-rank weighted risk: does covariance weighting lower the held-out weighted residual at matched rank?	**Q2.** Noise calibration: is the assumed covariance calibrated on this data?	**Q3.** Named physical truth metric: does it agree with recorded truth?
CRESST-II/III pulse shapes	*Yes, positive.* EMPCA lower in all 7 channels; equal-channel mean 21.6%, hierarchical 95% CI $[10.3, 38.8]$%	*Not tested.* A stationary PSD with shrinkage 0.1 is assumed, not validated	– No recorded truth: residuals only
GWOSC O1 H1/L1	*Not the question asked.* Used as a calibration stress test	*Yes, negative.* Global and local-64 s PSD models fail the null-spread gate on both detectors (up to 10.5); implementation checks pass	– No injected reference
TIDMAD validation files	*Whole window: yes, in direction only* (band-EMPCA below band-PCA, margin small beside the gap to a bin read). **Narrowband: no** — -3.8×10^{-5} $[-2.3 \times 10^{-4}, +1.3 \times 10^{-4}]$, the registered direction not met	*Yes, positive, second moments only.* χ^2/dof 1.0031, score dispersion 1.0142 in $[0.9, 1.1]$; no tails, dependence or coverage	*Yes.* Injected reference channel; scored as squared injected-amplitude error in readout units. **Whole window: no useful recovery. Narrowband: below a bin read in direction, all three verdicts withheld** (dependence did not stabilise)

6.2 CRESST: within-channel reconstruction

The public CRESST-II/III pulse-shape release [19, 26] supplies seven run/channel pairs, each treated independently and split chronologically (70% calibration, 30% evaluation), with pulse and

noise streams split separately; no evaluation pulse entered a fitted object, and within-stream row and trace-hash overlap is zero. The pooled held-out comparison is 19,335 pulses. The frozen model is rank-6 EMPCA with a stationary PSD covariance and shrinkage 0.1; the learned models are affine ($\mu + P(x - \mu)$) while single-template OF is a line through zero.

The same-rank contrast is the covariance-weighting evidence. Holding rank at six, EMPCA had the lower channel-mean weighted residual in *all seven* channels, with fractional reduction $1 - R_{\mathrm{EMPCA}}/R_{\mathrm{PCA}}$ from 1.6% to 62.0% and an equal-channel mean of 21.6%; the hierarchical 95% interval for that estimand is $[10.3, 38.8]$% (channel-fixed $[17.4, 27.1]$%, ratio-of-means sensitivity $[12.1, 39.8]$%, block lengths 16–256 moving it by under 0.35 points), from a channel-then-event bootstrap with 10,000 replicates. All seven channel effects are individually positive with intervals excluding zero, and nearly all the uncertainty is channel-level. This is a *sensitivity analysis of the archive, not an independent replication*: seven channels from two runs are seven correlated clusters, one operating point, one rank, one shrinkage.

The signal-model contrast is capacity, not geometry. Rank-6 EMPCA also beat single-template OF in every channel, by an equal-channel fractional reduction of 96.2%, and met the multi-template competitiveness gate everywhere; that contrast is an order of magnitude larger than the same-rank one and primarily measures replacing a fixed template with a rank-6 model, so we keep the two apart. All five negative controls degraded at the pooled level, with two of seven wrong-PSD channel intervals including zero. The tied-AE row is a *parameterization-consistency check of two code paths*, not an independently optimized model; gate S3 is the independent optimizer check. Channel-level tables and every control are in Supplement, Section S16.

6.3 GWOSC: a calibration boundary

Public 4-kHz H1/L1 strain around GW150914 [20, 27] was used as a stress test, not an event search: one 256-second interval, 62 valid off-source windows per detector, a Hann-windowed bias-corrected median-Welch PSD from calibration windows, and template-score dispersion on held-out windows over five random and five chronological splits, with a null spread near one as the target.

The implementation checks passed and the calibration tests did not. PSD ratios against GWpy [28] were one at recorded precision and the shared-FIR paths agreed to $\sim 10^{-13}$; yet the global PSD failed the aggregate random and chronological gates for both detectors, with individual null-spread ratios reaching 10.480 (H1) and 7.619 (L1), and the predeclared 64-second local model passed its stationary synthetic control (1.148) but failed on real H1 and L1 (9.424, 8.827). Event scores and nominal injections remain archived diagnostics; they are not converted to significance, false-alarm probability or sensitivity.

This is a boundary result, not a verdict on the geometry, and it motivates the next stage: because GWOSC carries no injected reference, a calibration failure there cannot be separated from the question the theory asks. A structurally wrong covariance model, moreover, need not have a large ε and its ε does not shrink with more calibration data (Section 3.4).

6.4 TIDMAD: recovery against recorded truth at the whole window

The TIDMAD release from the ABRACADABRA axion search [21] records an injected reference channel alongside the SQUID readout at 10 MHz, so a reconstruction can be scored against a real injected signal in real detector noise. A redesigned protocol, predeclared and executed once, asked the recovery question on untouched data. Its endpoint is the physical injected amplitude in readout

units, $a = 2q/\sqrt{w}$ with q the unitary-DFT coefficient at the injection bin, so window lengths are comparable; its truth label transfers the reference coefficient to readout units through a complex gain fitted on 64 disjoint transfer-calibration windows per stretch; every stretch is characterized (line, drift, coupling, multi-tone) on those calibration windows *before any method is fitted*; and the usefulness criterion is predeclared — an estimator is useful only if it beats both the zero predictor and the calibration-mean predictor with 95% intervals excluding zero. Three development files selected $w = 65536$ at rank 2, three validation files confirmed the direction, and the final test ran once on fourteen files, of which eight were eligible-coupled.

There is no useful recovery at this operating point. Over the eleven eligible stretches rank-2 PCA and EMPCA are indistinguishable from predicting zero (mean amplitude RMSE 0.661 and 0.655 against 0.654) and about 2.7× worse than reading the injection bin with no denoising (0.246). The combined paired contrasts of EMPCA against both usefulness references include zero ($+1.6 \times 10^{-5}$ $[-8.9 \times 10^{-5}, +1.2 \times 10^{-4}]$ and $+4.6 \times 10^{-4}$ $[-3.8 \times 10^{-5}, +9.6 \times 10^{-4}]$), while the bin read beats EMPCA at -1.19×10^{-2} $[-1.31 \times 10^{-2}, -1.08 \times 10^{-2}]$. No "useful" verdict was issued anywhere — both because the criterion fails on its own terms and because the frozen block-dependence assessment was inconclusive at nine evaluation blocks, which by the declared rule withholds it.

Three further facts are recorded. EMPCA does beat PCA consistently, in the direction the metric predicts, placing 2.8×10^{-3} of its reconstruction energy at the injection bin against PCA's 2.1×10^{-4} — an order of magnitude more, still not enough to compete with the bin itself. The quadrature-OF and bin-read estimators agree to 2.6×10^{-13}, confirming the diagonal-Fourier-weighting identity on real data. And the precision-weighted estimate is *effectively one file*: `0019`, whose injected amplitudes are smallest, carries 98.65% of the inverse-variance weight while the equal-file mean is $+7.8 \times 10^{-3}$ (Fig. 3); two post-hoc sensitivities on the eight eligible files agree about scale and differ about sign, and with $I^2 \approx 0.98$ neither replaces the predeclared primary, which stands as executed.

The leading hypothesis for the cause is dimensional, and it is stated as a hypothesis. At $w = 65536$ the real-view dimension is $p = 65536$ against $n = 350$ fit windows, $p/n = 187$. In the spiked-covariance model a population spike must exceed the noise level by about $\sqrt{p/n} \approx 14$ before the leading sample eigenvector overlaps it, and the injected tone's bin power is about 25 times the noise floor — near that threshold rather than clearly beyond it, and the measured bin SNR is not a population spike in any case. What is measured is the outcome: the injection bin carries 1.65×10^{-3} of the window energy while PCA's rank-2 reconstruction places 2.1×10^{-4} there, an under-representation of about 8×. Whether estimation error in the whole-window subspace is the cause is what a narrowband representation at the same window length is designed to decide.

6.5 TIDMAD: a narrowband representation at the same window

The whole-window stage left one hypothesis: that the failure was estimation error in a 65 536-dimensional subspace fitted from 350 windows. T22 tests it by removing exactly that factor and keeping everything else — the same window length, endpoint, truth labels, transfer calibration, rank, baselines, bootstrap and reporting. Only the space the rank-2 subspace is fitted in changes: from the whole window to the $2B$ real coordinates of the B unitary-DFT bins centred on the injection bin.

Design, in two recorded freezes. A stage-0 protocol fixed the allocation, representation, estimators, selection rule, admissibility, eligibility branches, dependence policy, inference, multiplicity

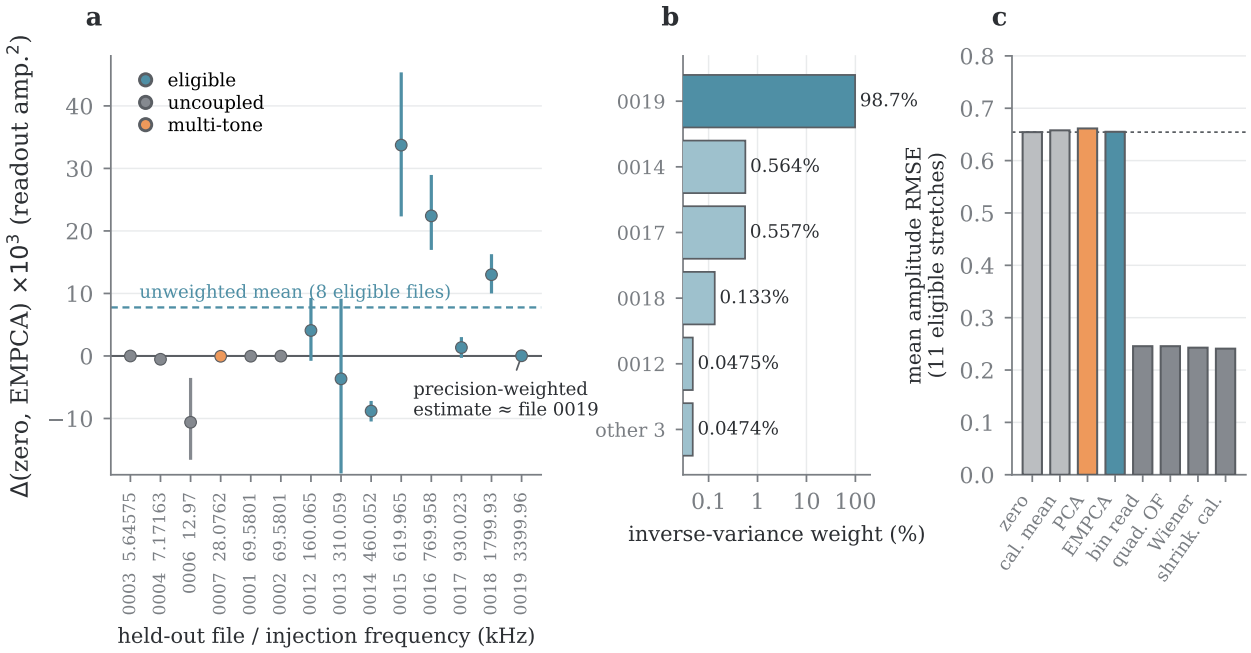


Figure 3: TIDMAD held-out operating-range test, per file ($w = 65536$, rank 2). (a) The paired contrast $\overline{\Delta}(\text{zero}, \text{EMPCA})$ on each of the fourteen final-test files, ordered by injection frequency, with its joint calibration/evaluation block-bootstrap 95% interval; a positive value favours EMPCA over predicting zero. The dashed line is the unweighted mean of the eight eligible files; the precision-weighted estimate coincides with file `0019`. (b) Inverse-variance weight shares. (c) Mean amplitude RMSE by method over the eleven eligible stretches: the learned rank-2 reconstructions sit at the trivial-predictor level (dotted), the frequency-domain estimators at 0.24–0.25. Exported at full precision from the archived stage outputs by `scripts/export_tidmad_heldout.py`; see `evidence/data/paper1/mlst_2026-09-11/README.md`.

and budget *before any code was written*; a selected-band predeclaration fixed B, the block lengths, the dependence verdict, the file roles and every code and input hash *before any hold-out outcome was scored*. Each stretch is 64 transfer-calibration + 350 fit + 300 evaluation windows with 16 384-sample guards, 46 825 472 samples in all. Hold-out stretches are *new time regions* of files already examined — starting at sample 37 027 840, which is the whole-window stretch end plus two guards, so there is no overlap — and this is a temporal hold-out, not independent-file replication. $B = 1$ is the identity control: a rank-2 projector on two real coordinates is the identity, so band-PCA = band-EMPCA = bin read, asserted to a relative 10^{-12} on all six development stretches and measured between -1.2×10^{-18} and $+2.9 \times 10^{-17}$. $B = 9$ was selected on development by the declared rule — the smallest admissible candidate whose equal-stretch mean is within one paired-bootstrap standard deviation of the best (Fig. 4c) — giving $p/n = 18/350 = 0.051$ against the whole window's 187.

The dimensional hypothesis is confirmed in direction. Over the seven eligible stretches the mean amplitude RMSE of the learned rank-2 reconstruction falls from 0.655 to 0.249, while the zero predictor stays at 0.654–0.689 and the bin read at 0.246–0.261 (Fig. 4a). The learned reconstruction moves from *indistinguishable from predicting zero and* 2.7× *worse than a bin read* to *better than the bin read*. On the frozen final test the equal-file contrast $\overline{\Delta}(\text{bin read}, \text{band-EMPCA})$ is $+1.58 \times 10^{-3}$ $[+3.5 \times 10^{-4}, +3.2 \times 10^{-3}]$, positive on all four eligible files with each file's own interval excluding zero, and the validation stage gave the same direction ($+1.82 \times 10^{-2}$ $[+2.5 \times 10^{-3}, +3.1 \times 10^{-2}]$).

All three verdicts are nevertheless inconclusive. The frozen dependence assessment did not stabilise — the aggregate block-variance curve is still rising at the largest usable block, $V(32)/V(16) = 1.745$, leaving nine complete blocks, the *same* limiting count as the whole-window stage's 150/16 — and the rule declared before development withholds every benefit verdict in that case, not only the limited-recovery one. Table 3 reports the three claims with what their estimates and intervals would have said. Two would have been declared (limited recovery, Holm-adjusted $p = 0.020$; denoising benefit, $p = 6.0 \times 10^{-4}$) and one would not.

The metric contribution is absent, and there is a structural reason. The combined contrast is $\overline{\Delta}(\text{band-PCA}, \text{band-EMPCA}) = -3.8 \times 10^{-5}$, with interval $[-2.3 \times 10^{-4}, +1.3 \times 10^{-4}]$ and $p = 0.70$: the point estimate marginally favours band-PCA, and the registered direction is *not met*. The explanation is post hoc but structural: band-PCA and band-EMPCA differ only in the metric, and inside a nine-bin band the noise PSD is nearly flat, so the whitening weight is nearly proportional to the identity. The measured in-band dynamic range $\max_k N_k / \min_k N_k$ is 1.16–4.66 across the eight hold-out stretches, against 4.77×10^4 over the interior bins of the whole window — four orders of magnitude less for the metric to act on. This is the interesting negative result of the stage: **the same restriction that fixes the estimation problem removes most of the metric's leverage**, so the paper's two determinants trade off against each other here. Whether a band wide enough to carry PSD structure and narrow enough to keep p/n small exists on this instrument is untested; the four candidates do not span it.

What else the run records. File `0016` was eligible-coupled in the whole-window time region and is *chirped* in the new one (drift 5.54 bins against the declared 0.5); by the declared rule it is evaluated, reported and excluded, leaving four eligible final-test files — exactly the declared

Table 3: T22 registered verdicts, frozen final test, $B = 9$, four eligible hold-out files. The estimand is the equal-file mean of the per-file paired squared-error contrasts, with a two-stage bootstrap interval (10 000 replicates) that carries within-file calibration and block uncertainty; p is two-sided from that bootstrap and Holm-adjusted across the three claims at familywise 0.05. **Every verdict is withheld** because the frozen dependence assessment did not stabilise — the rule was declared before development. Positive favours band-EMPCA.

claim	contrast	estimate	95 % interval	Holm p	verdict
limited recovery	$\overline{\Delta}$(zero, band-EMPCA)	$+0.3927$	$[+0.0045, +0.798]$	0.020	inconclusive
	$\overline{\Delta}$(cal-mean, band-EMPCA)	$+0.3935$	$[+0.0046, +0.798]$		
denoising benefit	$\overline{\Delta}$(bin read, band-EMPCA)	$+1.58 \times 10^{-3}$	$[+3.5 \times 10^{-4}, +3.2 \times 10^{-3}]$	6.0×10^{-4}	inconclusive
metric contribution	$\overline{\Delta}$(band-PCA, band-EMPCA)	-3.8×10^{-5}	$[-2.3 \times 10^{-4}, +1.3 \times 10^{-4}]$	0.70	inconclusive

threshold below which a combined interval would have been labelled indicative only. The inverse-variance combination is again dominated by one file ($I^2 = 0.986$), which is why the equal-file mean is the primary estimand; the equal-file t-interval on the point estimates alone includes zero on every contrast while the two-stage bootstrap excludes it on three, and both are reported. `quadrature-OF` reproduces the bin read to 3.8×10^{-12} on every eligible stretch. 240 of a declared cap of 250 fits were spent. The full report, including the one amendment made on development material and the deterministic re-score that followed it, is archived with the run.

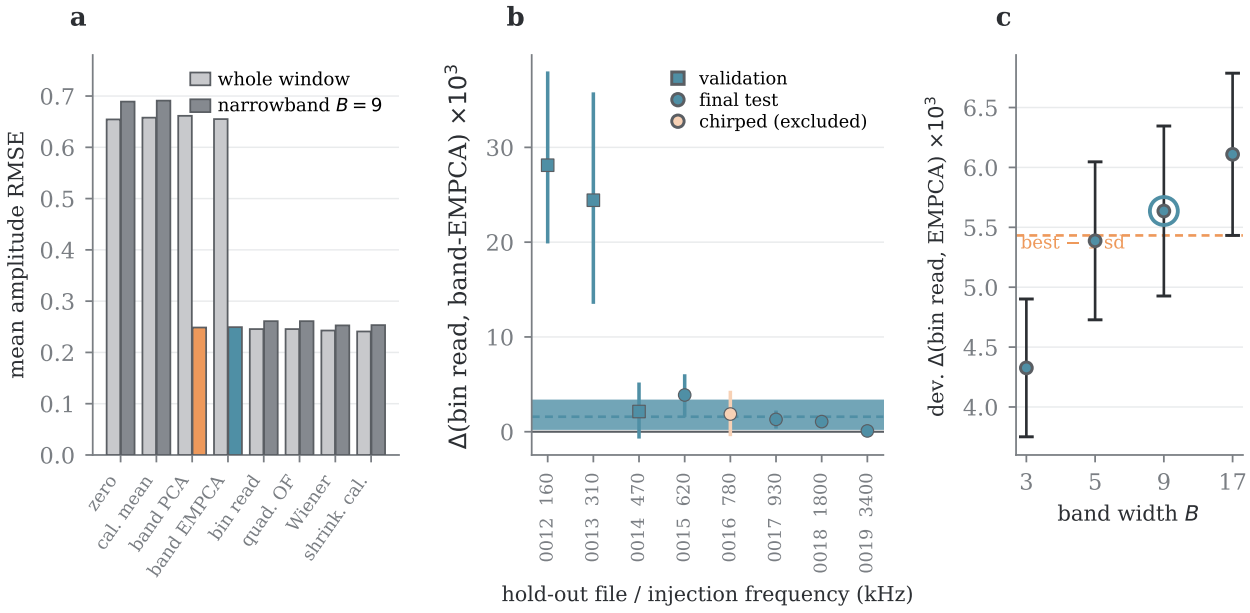


Figure 4: T22 narrowband hold-out at $w = 65\,536$, rank 2, $B = 9$. (a) Mean amplitude RMSE by method: the whole-window stage (light, eleven eligible stretches) against the narrowband stage (dark, seven eligible stretches). The learned rank-2 reconstructions move from the trivial-predictor level to below the bin read; the zero predictor and the frequency-domain estimators do not move. (b) Per-file $\overline{\Delta}(\text{bin read}, \text{band-EMPCA})$ with joint calibration/evaluation block-bootstrap intervals, ordered by injection frequency; squares are validation, circles the frozen final test, the pale marker is the chirped file excluded by the declared rule. The band and dashed line are the equal-file combined estimate and its two-stage bootstrap interval over the four eligible final-test files. (c) Development band selection: equal-stretch mean with paired-bootstrap standard deviation per candidate, the "best minus one sd" threshold, and the selected $B = 9$ (circled). Registered inputs and field names: `generated/SOURCE_MAP_part2.md`.

7 Discussion

Under the stated linear Gaussian model a reconstruction method is the pair $(\Sigma^{-1}, \mathcal{F})$ of a residual metric fixed by the measured noise and an admissible class fixed by the architecture. Once the metric is fixed, the four classes here are not competing likelihoods but different feasible sets for one Σ^{-1}-orthogonal projection, and the separable class is the first at which the architecture rather than the metric decides what can be represented. What this version adds is what happens to that statement when the covariance is estimated and the sample is finite, and the answers are more specific — and more negative — than the identities alone suggest.

The metric is only as good as the covariance supplied, and sensitivity is not a function of the error's size. Proposition 3.2 locates it in the coupling block of the relative error between retained and discarded eigenspaces, divided by the pairwise gaps, and the controlled study found the three regimes it separates: a first-order regime in which the displacement follows the coupling block within a factor of two; an ordering regime in which a diagonal error of any size leaves the subspace untouched until the spectrum reorders and then exchanges it; and a conditioning regime — near-singular empirical covariance — that no perturbation statement here covers and that shrinkage repairs only in its spectrum. That is why two perturbations of equal relative norm differed by two orders of magnitude in effect, and why the global norm bound, valid wherever its hypothesis held, was uninformative there. A finite-perturbation statement of the same shape remains open.

The class trades approximation for estimation, measurably but not predictably. Proposition 3.3 separates the finite-sample class contrast into a provably positive population approximation cost and two estimation excesses, and the accounting observed both regimes on one configuration, with the crossover bracketed between $N = 800$ and $N = 1600$ on fresh seeds. The registered c/N scaling was *refuted*, its forecast crossover wrong by a factor of six, and the interpolated $N^\star \approx 960$ is interpolation, not measurement. Nor does excitation in training make a direction recoverable: all five registered predictions of the excitation study failed, and the exploratory spectrum-and-loadings reading is a hypothesis for an independent test that supplies no operational threshold.

On real data the boundary is where the evidence is, and the two determinants trade off. CRESST gives a positive same-rank result under an assumed covariance it cannot validate; GWOSC shows that the assumed covariance can fail calibration badly while every implementation check passes; TIDMAD supplies recorded truth and reports no useful whole-window recovery, with a dimension-to-sample ratio of 187 as the leading hypothesis. Testing that hypothesis directly — restricting the same rank-2 fit to a nine-bin band at the same window — moved the learned reconstruction from the level of predicting zero to below a no-denoising bin read, and then produced two results the design did not anticipate. Its block-dependence assessment did not stabilise, leaving nine complete blocks, the same limiting count as the whole-window stage, so the rule declared before development withheld all three benefit verdicts. And the metric's contribution disappeared: inside a nine-bin band the noise PSD is nearly flat, so band-PCA and band-EMPCA nearly coincide. That is the sharpest statement this paper can make about the interaction of its determinants — **on this instrument the restriction that repairs the estimation problem removes most of the geometry's leverage**, and a band that does both is not among the four tested.

Untested design suggestions. Two follow from Section 4.1 and are recorded as suggestions, not results: prewhiten before tokenization, and tokenize along the axis in which Σ^{-1} is diagonal. Whitening supplies no locality, no graph structure and no deformation stability; those are properties of the detector domain, not of the noise model.

8 Limitations

Scope. The identities concern linear subspaces and Gaussian noise with a known or independently estimated positive-definite covariance. They do not cover curved signal manifolds, saturation, pile-up, heavy-tailed contamination, or a covariance estimated from the event being reconstructed. Timing variation can be approximated locally by extra linear directions, but a finite subspace need not represent large shifts. A Mahalanobis objective is efficient only when its noise model is adequate: covariance drift or outliers can invalidate its uncertainty interpretation even when the optimization is correct.

Identifiability. An eigendecomposition of a covariance produces decorrelated statistical modes, not uniquely labelled physical sources. Nothing here recovers a physical noise decomposition from a measured total spectrum, and no result attributes residual improvement to a named noise mechanism.

CRESST. Seven run/channel clusters from two runs, one operating point, one rank, one shrinkage. The primary predeclared interval resamples 19 335 pooled events and quantifies event-level uncertainty *conditional on these channels*; it does not quantify variation from sampling a new channel. The post-freeze channel-then-event analysis retains the effect directions but its intervals are coarse with seven clusters, and one negative-control exception (wrong-PSD weighting on `35/58.0`) shows that pooled control behaviour need not hold per channel. The trigger-efficiency calculation was underpowered over the frozen amplitude grid — six channels inconclusive at the primary 5% false-positive rate, none reaching 90% efficiency — so reconstruction gains cannot be translated into a threshold, an exposure or a sensitivity improvement. Rank/shrinkage robustness and time-cut sensitivity are *deferred*, not established.

GWOSC. One 256-second interval, one event region, two PSD models. The shared-FIR result says two software paths agree when they implement the same finite-duration statistic; it does not establish equivalence between the PSD-domain GLS statistic and a finite FIR one, and agreement with GWpy cannot make a Gaussian PSD model adequate for nonstationary real noise.

TIDMAD. Named validation files at named operating points, not a survey. The whole-window stage established *no* useful recovery and did not confirm the rank-2 prediction on any real file; the linear estimator is not competitive on the release's own detector-facing denoising score. The narrowband stage is a *temporal hold-out* on files already examined — new time regions of previously used files — and is not independent-file replication; its dependence assessment leaves nine complete blocks at the largest block length, and the declared rule withholds every benefit verdict when that assessment does not stabilise. Four eligible final-test files is a small number, and the equal-file t-interval on the point estimates alone includes zero on every contrast. The calibration model is checked at second moments only: no tails, no dependence, no coverage. The flat-in-band explanation of the vanished metric contribution is post hoc.

Controlled studies. One generator per study, oracle or engineered covariance, fixed rank. The covariance study is conditional on a fixed training draw with engineered perturbations — a local theorem check, not a finite-perturbation guarantee, and silent about an independently estimated covariance in deployment. The structural-prior accounting uses a Monte-Carlo covariance proxy, and its restricted reference is the best of 51 alternating starts: restart agreement, not a certified global optimum. Restart spreads are of the order of the contrast near the sign change, so the optimization term is visible exactly where the classes are closest. The excitation study's spectrum-and-loadings reading is post hoc, checked on the runs that suggested it. The tied-versus-untied ablation and the July excited-support experiment belong to the held evidence (Section 1) and are

not reported here.

Not claimed anywhere. Detection sensitivity, energy resolution, false-alarm probability, a universal recovery threshold, an operational abstention rule, exact restricted population optima, or a validated c/N law.

9 Conclusion

Under a linear Gaussian measurement model the inverse noise covariance is the metric in which a detector reconstruction is learned; the architecture fixes only the admissible class, and the training data only the excited directions. Optimal filtering, covariance-weighted PCA, tied noise-aware linear autoencoders and their Kronecker-separable restriction are one projection in that metric, and the residual it minimizes predicts the whitened, not the Euclidean, clean-signal error. At finite sample the metric's sensitivity to covariance error is local, with a separate conditioning regime no perturbation bound covers; the class's structural prior trades a provably positive approximation cost against an estimation saving whose crossover we bracket but cannot predict; and excitation in training is not detectability. On public detector data the weighting lowers every held-out weighted residual on seven CRESST channels, a Gaussian PSD model fails calibration on GWOSC while every implementation check passes, and against recorded TIDMAD truth no rank-2 whole-window reconstruction beats predicting zero — while the narrowband restriction that repairs that failure removes the geometry's leverage along with it. The determinants are separable, and each of them can fail on its own.

10 Code and data availability

All results come from public datasets; no internal or unreleased detector data, noise budget or instrument simulation is used or described. The CRESST pulse-shape release is described in Ref. [19] and hosted by the ORIGINS Dark Matter Data Center; the collaboration asks that the description and Ref. [26] be cited, and we do so. The GWOSC strain is the O1 bulk release [20, 27], CC BY 4.0; this research has made use of data obtained from the Gravitational Wave Open Science Center, a service of the LIGO Scientific Collaboration, the Virgo Collaboration and KAGRA. The TIDMAD data are the public release under CC-BY-4.0 [21].

Every quantitative claim is listed in the accompanying claim-to-evidence table with its mathematical assumption, dataset and split, output field, code revision, archived output path and limitation; its paths and a set of concrete values are machine-checked. The archived evidence is versioned with a complete SHA-256 manifest verified by one command; figures and generated tables are rebuilt by scripts that read every value from a registered file and hard-code none, and a source map ties each output cell to its file and field. Preregistrations, protocol JSONs, fit caps and the deviations decided before each evaluation are archived alongside, together with the per-file bootstrap draw arrays needed to recompute every interval and p-value without the instrument data. Supplement, Section S17 lists the archive identifiers, run commits and manifest digests.

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